

- Task-1

1. What Is AI ?

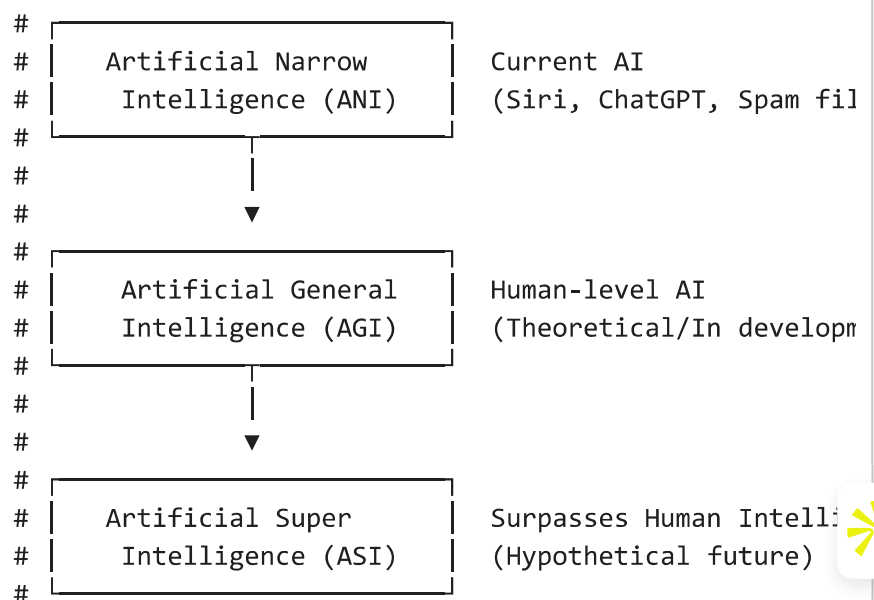
```

"""Artificial Intelligence (AI) refers to the simulation of human intelligence
learn, reason, and solve problems like humans.
Instead of following rigid pre-coded rules, AI systems analyze data, detect pat
and perform complex tasks such as language translation, decision-making, and in
"""

```

'Artificial Intelligence (AI) refers to the simulation of human intelligence in machines programmed to think,\nlearn, reason, and solve problems like humans.\n\nInstead of following rigid pre-coded rules, AI systems analyze data, detect p
atterns, adapt to new inputs, \nand perform complex tasks such as language tran
slation, decision-making, and image recognition.\n'

2. Types of Artificial Intelligence



Artificial Narrow Intelligence (ANI / Weak AI):

Designed to perform a specific, single task (e.g.,
All current operational AI systems belong to this c

Artificial General Intelligence (AGI / Strong AI):

A theoretical form of AI where a machine possesses and solving diverse tasks across different domains

Artificial Superintelligence (ASI):

A hypothetical future AI that surpasses human intel
emotional reasoning, and scientific knowledge.

"""

4. Real-Life Applications of AI

"""

Healthcare: AI-powered diagnostic tools analyze medical scans to
detect tumors earlier than traditional methods.

Finance: Fraud detection systems analyze transaction patterns in real-time to
prevent unauthorized purchases.

E-Commerce & Entertainment: Recommendation engines on Netflix, Spotify,
and Amazon suggest products/content tailored to use

Transportation: Navigation apps (Google Maps) predict real-time traffic flow ar
optimize delivery routes.

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ailored to user habits.\n\nTransportation: Navigation apps (Google Maps) predic
t real-time traffic flow and \n          optimize delivery routes.\n'
```

5. Advantages and Limitations of AI

#Advantages...

#1. 24/7 Availability: AI operates continuously without fatigue or drop in effi

#2. Reduction of Human Error: High precision in repetitive data-entry and mathe

#3. Fast Data Processing: Processes millions of data points per second to provi

Limitations...

1. High Cost of Implementation: Training complex models requires massive comp

2. Lack of Genuine Creativity & Emotion: Operates strictly on mathematical pr

3. Job Displacement Concerns: Automation of repetitive tasks can lead to work



Task 2: Setting Up AI Development Environment

"""

1. Step-by-Step Installation Guide

Install Python:

Download Python from python.org.

Run the installer and check the box: "Add Python to PATH".

Install Visual Studio Code (VS Code):

Download and install from code.visualstudio.com.

Open VS Code, navigate to Extensions (Ctrl+Shift+X), and install the Python Ext

Install Jupyter Notebook & AI Libraries:

Open your Terminal (Mac/Linux) or Command Prompt (Windows) and run:

"""

#Code Write in Command page:- `pip install jupyter numpy pandas`

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"""

2. Verification Script (hello_ai.py)

Create a file named `hello_ai.py` in VS Code or run it in a Jupyter Notebook cell

CODE(Input...):-

`print("Welcome to Artificial Intelligence")`

Output:

Welcome to Artificial Intelligence



📸 Screenshot Guide for Your PDF Report

Be sure to capture clear screenshots of the following 4 screens to attach under

Python Installation: Terminal showing `python --version`.

VS Code Environment: VS Code editor with `hello_ai.py` open.



Jupyter Notebook Interface: A running .ipynb notebook cell showing the executio

Terminal Output: Terminal window displaying the executed print output.
 ""

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

✓ Task 3: Python for AI

Program 1: Student Pass/Fail Predictor (Classification AI Concept)

```
# Pass/Fail Predictor
student_name = input("Enter student name: ")
marks = float(input("Enter exam score (0-100): "))

# Decision logic using operators
passing_score = 50.0

if marks >= passing_score:
    status = "PASSED"
    confidence = (marks / 100) * 100
else:
    status = "FAILED"
    confidence = ((100 - marks) / 100) * 100

print(f"\n[AI Result] Student: {student_name}")
print(f"Status: {status} (Model Confidence: {confidence:.1f}%)")
```

Enter student name: Samarth Modi
 Enter exam score (0-100): 87.83

[AI Result] Student: Samarth Modi
 Status: PASSED (Model Confidence: 87.8%)



Program 2: Weather AI Temperature Converter

```
# Temperature Converter for Weather AI
celsius_input = float(input("Enter temperature in Celsius: "))

# Conversion operators
fahrenheit = (celsius_input * 9/5) + 32
kelvin = celsius_input + 273.15
```

```
print("\n--- Weather AI Data Normalization ---")
print(f"Celsius:      {celsius_input:.2f} °C")
print(f"Fahrenheit: {fahrenheit:.2f} °F")
print(f"Kelvin:       {kelvin:.2f} K")
```

Enter temperature in Celsius: 35.9

```
--- Weather AI Data Normalization ---
Celsius:      35.90 °C
Fahrenheit: 96.62 °F
Kelvin:       309.05 K
```

Program 3: AI Service Age Eligibility Checker

```
# Age Eligibility Checker
user_age = int(input("Enter your age: "))

MIN_AGE_BASIC = 13
MIN_AGE_ADVANCED = 18

if user_age >= MIN_AGE_ADVANCED:
    access_level = "Full Access (Advanced AI Models & Generative Tools)"
elif user_age >= MIN_AGE_BASIC:
    access_level = "Restricted Access (Basic AI Tools with Safety Filters)"
else:
    access_level = "Access Denied (User below minimum age limit)"

print(f"\nAge: {user_age}")
print(f"Eligibility Result: {access_level}")
```

Enter your age: 21

```
Age: 21
Eligibility Result: Full Access (Advanced AI Models & Generative Tools)
```

Task 4: AI Case Study — Self-Driving Cars

""""

Overview & How AI is Used:-

Self-driving cars (such as Tesla, Waymo, and Cruise) leverage Artificial Intelligence to navigate roads safely without human intervention. The vehicle acts as an autonomous system that makes real-time driving decisions, and executes physical controls (steering, braking, and acceleration).

1. Input (Perception Layer)

Autonomous vehicles continuously gather high-dimensional sensory data from multiple sources, including cameras, radar, and LiDAR.

LiDAR (Light Detection and Ranging): Fires laser beams to construct precise 3D maps of the environment.



Cameras: High-resolution cameras capture visual data, including lane lines, traffic lights, and road signs.

Radar & Ultrasonic Sensors: Measure relative velocity and distance of surrounding objects under poor visibility conditions (rain, fog).

GPS & IMU: Provide location tracking and vehicle speed.

2. Processing (Cognition Layer)

The gathered data is processed by onboard edge-computing systems running Deep Learning algorithms.

Computer Vision (Convolutional Neural Networks - CNNs): Detects and classifies objects in the environment.

Sensor Fusion: Combines raw inputs from LiDAR, Radar, and Cameras to create a unified, accurate representation of the environment.

Path Planning Algorithms: Uses Reinforcement Learning and search algorithms (e.g., A* algorithm) to calculate optimal trajectories while anticipating movements of other vehicles.

3. Output (Action Layer)

The output of the processing system translates into physical vehicle commands sent to the vehicle's actuators.

Steering angle adjustments to stay centered in lanes or turn.

Brake pressure application when obstacles or traffic signals are detected.

Throttle adjustment to maintain speed limits or safely overtake cars.

4. Key Benefits

Safety: Over 90% of traffic accidents are caused by human error (distracted driving, impaired judgment). Autonomous vehicles eliminate human factors.

Accessibility: Provides independent mobility for the elderly and individuals with disabilities.

Traffic Optimization: AI-driven vehicles communicate to streamline traffic flow, reducing congestion and travel time.

Overview & How AI is Used:- Self-driving cars (such as Tesla, Waymo, and Cruise) leverage Artificial Intelligence to navigate roads safely without human intervention. The vehicle acts as an autonomous agent that perceives its environment, makes real-time driving decisions, and executes physical controls (steering, braking, and acceleration).

1. Input (Perception Layer) Autonomous vehicles continuously gather high-dimensional sensory data from multiple hardware sources:

- LiDAR (Light Detection and Ranging):** Fires laser beams to construct precise 3D spatial maps of the surroundings.
- Cameras:** High-resolution cameras capture visual data, including lane lines, traffic lights, and road signs.
- Radar & Ultrasonic Sensors:** Measure relative velocity and distance of surrounding objects under poor visibility conditions (rain, fog).
- GPS & IMU:** Provide location tracking and vehicle speed.

2. Processing (Cognition Layer) The gathered data is processed by onboard edge-computing systems running Deep Learning algorithms.

Task 5: Mini AI Project — Rule-Based Chatbot

```
def rule_based_chatbot():
    print("=====")
    print("🤖 Welcome to SimpleAI Chatbot 🤖")
    print("    (Type 'bye' or 'exit' to end conversation)    ")
    print("=====\\n")

    bot_name = "Botty"

    while True:
        # Get user input and normalize text to lowercase
        user_input = input("You: ").strip().lower()

        # Exit condition
        if user_input in ["bye", "exit", "quit", "goodbye"]:
            print(f"{bot_name}: Goodbye! Have a great day ahead! 🍌")
            break

        # Conversation Rules (if-elif-else logic)
        elif user_input in ["hello", "hi", "hey"]:
            print(f"{bot_name}: Hello! Welcome! How can I assist you today?")

        elif "how are you" in user_input:
            print(f"{bot_name}: I'm doing great, thank you for asking! I'm happy to help you.")

        elif "your name" in user_input or "who are you" in user_input:
            print(f"{bot_name}: My name is {bot_name}, your friendly rule-based chatbot.")

        elif "what is ai" in user_input or "define ai" in user_input:
            print(f"{bot_name}: AI stands for Artificial Intelligence—machine learning and deep learning.")

        elif "python" in user_input:
            print(f"{bot_name}: Python is the primary language used for building chatbots.")

        elif "fine" in user_input or "good" in user_input:
            print(f"{bot_name}: That's wonderful to hear!")

        elif user_input == "":
            print(f"{bot_name}: It seems you didn't type anything. Feel free to ask me anything!")

        else:
            print(f"{bot_name}: I'm sorry, I don't understand that yet.")

# Execute the chatbot
if __name__ == "__main__":
    rule_based_chatbot()
```

```
=====
🤖 Welcome to SimpleAI Chatbot 🤖
    (Type 'bye' or 'exit' to end conversation)
=====
```

You: Sammy



Botty: I'm sorry, I don't understand that yet. Try asking me 'What is AI?' or sa
You: hello
Botty: Hello! Welcome! How can I assist you today?
You: hi
Botty: Hello! Welcome! How can I assist you today?
You: good
Botty: That's wonderful to hear!
You: who are you
Botty: My name is Botty, your friendly rule-based assistant.
You: okay
Botty: I'm sorry, I don't understand that yet. Try asking me 'What is AI?' or sa
You: define ai
Botty: AI stands for Artificial Intelligence—machines programmed to think and le
You: bye
Botty: Goodbye! Have a great day ahead! 🤖

Start coding or [generate](#) with AI.

